

The Questions a Qualified Theory of Time Must Answer

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Abstract

If time is the measure of change, then any qualified theory of time must answer three fundamental questions: (1) What is changing? — the subject of change must be specified. (2) How does it change? — the dynamical driver of change must be identified. (3) What is the distinction between objective change and subjectively observed change? — the gap between reality and perception must be bridged. This paper argues that most existing philosophical and physical theories of time fail to meet these criteria. They define time circularly by clocks, leave the subject of change unspecified, offer unclear dynamical drivers, and conflate subjective observation with objective reality, thereby abandoning the pursuit of objective change in favor of an observer-dependent perspective.

Keywords: philosophy of time; ontology of time; criteria; change; dynamical driver; objective time; subjective time; operationalism

1. Introduction: The Unasked Questions

"Time is the measure of change."

This statement is almost universally accepted as intuitive. Yet, surprisingly, neither philosophy nor physics has systematically answered the three questions that immediately follow from it:

- (1) What is changing?
- (2) How does it change?
- (3) What is the difference between objective change and what we observe?

A theory of time that cannot answer these questions is incomplete. A theory that does not even ask them is inadequate. This paper proposes these three questions as the necessary and sufficient criteria for any qualified theory of time, and uses them to diagnose the shortcomings of existing approaches.

2. The Three Criteria

Criterion 1: What Is Changing? — The Subject of Change

If time measures change, the first duty of any time theory is to specify the entity or substance that undergoes change. Without a clearly identified subject, "change" is an empty word — change of what?

A qualified theory must answer:

- What is the fundamental ontology that persists through change?
- What is the variable or property whose variation constitutes change?
- Is the subject of change the universe as a whole, or only local subsystems?

Failure to meet this criterion results in change being attributed to vague notions like "events," "states," or "processes," without specifying what these are made of or what exactly is varying.

Criterion 2: How Does It Change? — The Dynamical Driver

Even if we know what changes, we must also know why and how it changes from one state to the next. A qualified theory must specify:

- What force, gradient, or principle drives the transition from one state to another?
- Is the dynamics deterministic, probabilistic, or a combination?
- Can the driver be expressed in terms of measurable physical quantities?

Failure to meet this criterion leaves change as a brute fact — things just happen to change, but we have no physical principle governing the transition. Time becomes a passive container rather than an active measure of real dynamics.

Criterion 3: Objective Change vs. Subjectively Observed Change

This is the most frequently neglected criterion. A qualified theory must draw a clear distinction between:

- **Objective change:** the actual transformation of the universe (or its subject) independent of any observer, clock, or measurement apparatus.
- **Subjective change:** the change perceived, measured, or sampled by an observer with finite sensory or instrumental resolution.

Failure to meet this criterion conflates epistemology with ontology — what we measure is mistaken for what is. The theory collapses into operationalism, abandoning objective reality in favor of observer-dependent readings.

3. Diagnosis of Existing Theories

3.1 Newtonian Absolute Time

What changes? Unspecified. Newton posits that time "flows equably," but does not specify what is flowing or what change time measures.

How does it change? Circular. "Flow" is just another word for time — he defines time using time.

Objective vs. subjective? Undistinguished. Absolute time is assumed to be objective, but no mechanism links it to observed time.

Newton gives us a coordinate for time, but not a definition of what time measures.

3.2 Leibnizian Relational Time

What changes? Vague. Time is the "order of events," but "events" are not ontologically defined.

How does it change? Unspecified. The ordering principle is logical, not physical.

Objective vs. subjective? Conflated. The order of events is defined relative to observers or reference frames.

Leibniz avoids circularity but replaces it with vagueness.

3.3 Operationalism (Bridgman)

What changes? Avoided. Time is "what clocks measure" — the subject of change is replaced by instrument readings.

How does it change? Circular. Clocks define time, time defines clocks.

Objective vs. subjective? Collapsed. There is no objective time beyond what is measured.

Operationalism is a valid measurement strategy but a failed ontological definition. It answers "how do we measure?" instead of "what is real?"

3.4 Relativity Theory

What changes? Geometric but ontologically unclear. Spacetime curvature changes, but what is curving? Vacuum? Field? The ontology is not settled.

How does it change? Gravity-driven (Einstein field equations) — this is the strongest answer among existing theories.

Objective vs. subjective? Heavily observer-dependent. Simultaneity, duration, and order all depend on the observer's motion. Objective time (if any) is obscured.

Relativity offers a brilliant kinematics of time but refuses to commit to an ontology.

3.5 Quantum Mechanics

What changes? The wavefunction. But the wavefunction's ontology is debated (real? epistemic? statistical?).

How does it change? Schrödinger equation — but this assumes time as an external parameter, not derived from dynamics.

Objective vs. subjective? Measurement problem. The collapse of the wavefunction introduces observer-dependence at the fundamental level.

Quantum mechanics treats time as a background parameter, not as something that emerges from change itself.

3.6 The "Time from Information" Approach

What changes? Information. But information is not a fundamental physical substance — it is always information about something.

How does it change? Information imprinting and accumulation. But the driver of imprinting is not specified.

Objective vs. subjective? Partially distinguished. But "information" is inherently epistemic — it requires a knower.

Information-based theories risk replacing circularity with abstraction. "Information" is a relational property, not an ontological substance.

4. Summary Diagnosis

Theory	Subject of Change Specified?	Dynamical Driver Specified?	Objective/Subjective Distinguished?
Newton	✗ No	✗ Circular	✗ No
Leibniz	✗ Vague	✗ No	✗ No
Operationalism	✗ Avoided	✗ Circular	✗ Collapsed
Relativity	✗ Unclear	✓ Yes (gravity)	✗ Observer-dependent
Quantum Mechanics	✗ Debated	Assumed (external t)	✗ Measurement problem

Theory	Subject of Change Specified?	Dynamical Driver Specified?	Objective/Subjective Distinguished?
Information Theory	✗ Abstract	✗ Unclear	Partial

None of the major theories satisfies all three criteria.

5. The Consequence of Failing These Criteria

When a theory of time fails to specify what changes, it abandons ontology.

When it fails to specify how change is driven, it abandons causality.

When it fails to distinguish objective change from subjective observation, it abandons reality in favor of perspective.

The cumulative effect is that modern physics and philosophy have, for centuries, been describing our measurement of time rather than time itself. We have become experts in clock-reading and amateurs in ontology.

This is not a failure of physics as a predictive science — it performs remarkably well. It is a failure of understanding. We can predict time with extraordinary precision, yet we cannot say what it is.

6. What a Qualified Theory Must Look Like

Based on the three criteria, a qualified theory of time should:

- (1) Specify the subject of change in ontologically unambiguous terms — e.g., the universe's energy distribution over a fixed spatial topology, rather than vague "events" or "states."
- (2) Identify the dynamical driver as a physically measurable principle — e.g., energy gradients drive redistribution, translating into well-known equations (diffusion, heat, Einstein field equations).

(3) Draw a clear line between objective and subjective time — e.g., global snapshot sequence (objective) vs. local, finite-sampling perception of that sequence (subjective).

This structure avoids circularity, specifies change, and gives a physical account of dynamics while preserving the distinction between what is and what we observe.

7. Conclusion

If time is the measure of change, then a theory of time must answer:

(1) What changes?

(2) How does it change?

(3) What is the difference between objective change and observed change?

Most existing theories fail on at least two of these counts. They define time by clocks, avoid ontology, obscure dynamics, or collapse objectivity into observation.

A qualified theory must do better. It must specify the subject, identify the driver, and distinguish reality from measurement. These are not optional extras — they are the minimum standards for any serious account of time.

Without meeting these criteria, any statement about time remains, at best, a useful operational tool, and at worst, a sophisticated evasion of the question it claims to answer.

This paper proposes a diagnostic framework rather than a complete theory. The author's own positive account — time as the sequence of cosmic energy-distribution snapshots, driven by energy gradients — is developed elsewhere and satisfies all three criteria outlined here.

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